# Introduction to Panel Data Analysis

*These notes accompany Lecture 12. They combine the slide material with the explanations given in class. Terms printed in bold are defined in the glossary at the end.*

Panel data follow the same units over time, and that single feature solves a problem no cross-section can: it lets us control for things we never observe. The running example is a wage equation for 545 men observed annually from 1980 to 1987.

**Learning outcomes – after this lecture you should be able to:**

1. Explain what panel data are and why they allow control of unobserved heterogeneity.
2. Write the unobserved effects model and distinguish one-way from two-way error components.
3. Derive the within transformation and explain why time-constant regressors drop out.
4. Compare pooled OLS, fixed effects and random effects on assumptions, consistency and efficiency.
5. Use the LM and Hausman tests to choose among the three.
6. Combine fixed effects with instrumental variables (FE-2SLS).
7. Set up a difference-in-differences estimator and state the parallel trends assumption.

## Contents

1. Why panel data?
2. The unobserved effects model
3. Pooled OLS
4. Fixed effects estimation
5. FE-2SLS estimation
6. Random effects estimation
7. Model diagnostics and choosing an estimator
8. Difference-in-differences

Appendix. The correlated random effects approach

# 1. Why panel data?

A **panel** follows the same N units – people, firms, banks, routes, countries – over T time periods, giving N × T observations. Panel data carry both subscripts:

$$Y_{it} \quad \textit{for } t = 1, \ldots, T \textit{ and } i = 1, \ldots, N$$

The key distinction from a pooled cross section is that the same units are followed over time. A household survey repeated annually with different households each year is a pooled cross section; a true panel would let you follow each household across periods.

If every unit is observed in every period the panel is balanced. Researchers often work with unbalanced panels, where different units have different numbers of observations. Because panel data are sometimes stored with all periods for one unit stacked, then the next unit, they are also called longitudinal data.

## 1.1 Advantages

- **Control of unobserved heterogeneity.** Ability, managerial quality, route attractiveness – fixed traits we never observe. Multiple observations on the same unit let us control for them, so the omitted-variable problem that would bias a single cross-sectional regression need not arise here.
- **Sample size.** Pooling raises N × T considerably, so much better estimators can be obtained. This matters especially for annual macro data, where few series extend beyond fifty years and it is hard to obtain significant t-ratios from a pure time series.
- **Micro-level detail.** Panels are most often collected on individuals, households and firms, avoiding the aggregation problem.

## 1.2 Limitations

- Harder to collect. Following the same units is expensive, and non-response compounds across waves.
- Gaps and dropouts are the norm. Firms fail, people leave the survey, a bank is absorbed. Real panels are usually uneven – and the missing pieces often are not missing at random.
- Measurement error, from imprecise questions, poorly conducted surveys, and respondents' memory.
- Too few periods or too few units, and the methods below do not behave well.

## 1.3 Setting up a panel in EViews

1. Organise the data in Excel with a column identifying the cross-section, a column identifying the time period, and observations arranged by period within each unit – long format.
2. Import into EViews, specifying the two identifiers correctly. The workfile then becomes a panel workfile.

## 2. The unobserved effects model

Variables get omitted from a model for two reasons: analyst error, and missing data. The first is easier to correct. Data may be missing because they were not collected for this study, or because they cannot be collected for methodological reasons. Either way, certain characteristics of the units remain unobserved.

With one observed regressor, the model is:

$$y_{it} = \beta_0 + \beta_1 x_{it} + a_i + u_{it}$$

| Term | Name | Meaning |
|---|---|---|
| $a_i$ | Unobserved effect / individual heterogeneity / fixed effect | All unobserved, time-constant factors affecting y. The absence of a t subscript is the point – it does not change over time |
| $u_{it}$ | Idiosyncratic error | Unobserved factors that change over time; like the errors in a time series regression |
| $v_{it} = a_i + u_{it}$ | Composite error | What pooled estimation leaves in the error term when $a_i$ is not modelled |

*The estimators that follow differ in how they treat $a_i$.*

### 2.1 One-way and two-way error components

$$\textit{One-way:}\quad v_{it} = a_i + u_{it} \qquad \textit{Two-way:}\quad v_{it} = a_i + \lambda_t + u_{it}$$

The one-way model separates a unit-specific term $a_i$ from noise. Because $a_i$ is shared by every observation on the same unit, two observations on that unit are correlated.

The two-way model adds a time effect $\lambda_t$, carrying only the t subscript, common to all units in a given period – aggregate influences such as macroeconomic shocks, the business cycle, or a nationwide policy change. Observations are then linked in two directions: same unit share $a_i$, same period share $\lambda_t$.

## 3. Pooled OLS

The simplest approach ignores $a_i$ entirely: stack all N × T observations and run OLS on the composite error. It estimates a common constant for all cross-sections, which amounts to assuming there are no differences among units – useful only if the data set really is homogeneous a priori.

| Strengths | Weaknesses |
|---|---|
| Simplest possible estimator; a natural baseline | Consistent ONLY if Cov(x, $a_i$) = 0 – the strong assumption we wanted to escape |
| Uses both within- and between-unit variation | $a_i$ sits in $v_{it}$ every period, so the composite errors are serially correlated within units |
| Estimates time-invariant regressors such as education and race without difficulty | Default standard errors are therefore wrong – you must cluster by entity |

*Note that the serial correlation arises by construction, not by accident: the same $a_i$ appears in every period for that unit. Even when pooled OLS is consistent, its standard errors are not usable untreated.*

# 4. Fixed effects estimation

The fixed effects method treats the constant as unit-specific, allowing a different intercept for each unit.

## 4.1 The within transformation

Take the model for a single unit, average it over time, and subtract:

$$y_{it} = \beta_1 x_{it} + a_i + u_{it}$$

$$\bar{y}_i = \beta_1 \bar{x}_i + a_i + \bar{u}_i$$

$$(y_{it} - \bar{y}_i) = \beta_1 (x_{it} - \bar{x}_i) + (u_{it} - \bar{u}_i)$$

Because $a_i$ is fixed over time it appears identically in both equations and cancels. Estimating the demeaned equation by pooled OLS gives the fixed effects – or within – estimator. The second name comes from the fact that it uses only the time variation in y and x within each unit.

Adding more regressors changes nothing: time-demean each one and run pooled OLS on all the demeaned variables.

## 4.2 Assumptions

- **Strict exogeneity, for consistency:** $E(u_{it} \mid x_{i1}, \ldots, x_iT, a_i) = 0$ for all t. The idiosyncratic error must be uncorrelated with each regressor in every period, not just the current one.
- **Homoskedastic, serially uncorrelated $u_{it}$, for efficiency.**

> **The crucial licence fixed effects gives you**
>
> No assumption at all is needed on Cov(x, $a_i$). Fixed effects allows arbitrary correlation between the unobserved effect and the regressors in any period.
>
> That is the whole reason for using panel data in the first place – and the reason FE is the default when in doubt.

## 4.3 Why time-constant variables disappear

Any regressor constant over time for all i is swept away: if $x_{it}$ does not vary with t, then $x_{it} - \bar{x}_i = 0$ for all i and t. So gender, a city's distance from a river, schooling, race and ethnicity cannot be estimated.

> **Experience is the subtle case**
>
> Experience is not time-constant – it rises by exactly one each year. But because it rises by exactly one for everyone, with both individual and time effects in the model it is perfectly collinear with them and drops out too.
>
> Experience-squared survives, because it is non-linear.

Time-constant variables can still enter as interactions with variables that do change. Interacting education with each year dummy, for instance, shows how the return to education has changed over time.

## 4.4 The wage equation: fixed effects versus pooled OLS

Data from Vella and Verbeek (1998): 545 men, each working every year from 1980 to 1987.

| Coefficient | Pooled OLS | Fixed effects | Reading |
|---|---|---|---|
| MARRIED | 0.108 | 0.047 | Roughly halves |
| UNION | 0.182 | 0.080 | Roughly halves |
| $R^2$ | 0.19 | 0.62 | Not directly comparable – see below |

Men who marry or unionise differ in stable, unobserved ways. The model without individual effects wrongly credits those fixed differences to marriage and union status. Fixed effects strips them out and identifies the premium only from how a given person's wage changes when his status changes.

### Two things not to over-read

The $R^2$ jump from 0.19 to 0.62 is not "three times the fit". The two models explain different things. What it does tell you is that individual effects account for a large share of wage differences.

EViews reports a common intercept because after the within estimation it recovers the individual effects and normalises them to sum to zero. So C is the average individual intercept – the baseline around which each unit's own effect is a deviation, and C + $\hat{a}_i$ gives unit i's own intercept.

## 4.5 Testing whether the effects belong

The null is that all individual effects are zero. The test compares the residual sums of squares from the pooled and FE models. Reject and the effects belong in the model; fail to reject and pooled OLS will do.

| Test | Statistic | Conclusion |
|---|---|---|
| Cross-section effects | F = 9.16, df (544, 3805) | Reject – keep individual effects |
| Period effects | F = 29.24, df (7, 3805) | Reject – keep time effects |
| Both jointly | F = 10.07, df (551, 3805) | Reject – the two-way specification is justified |

## 4.6 Three equivalent routes to the same estimator

| Method | How $a_i$ is removed | Identification comes from |
|---|---|---|
| Within transformation | Subtract each unit's own time-average | How each observation deviates from that unit's mean |
| First differencing | Subtract the previous period; $a_i - a_i = 0$ | Period-to-period changes. Loses the first period |
| Dummy-variable regression (LSDV) | Not removed – estimated, one intercept per unit | Identical $\beta_1$ and standard errors to time-demeaning. Infeasible for large N |

### Choosing between FE and first differencing

When the $u_{it}$ are serially uncorrelated, fixed effects is more efficient.

If $u_{it}$ follows a random walk, then $\Delta u_{it}$ is serially uncorrelated and first differencing is better. When they give substantively different results the choice is genuinely hard. Report both and try to work out why they differ.

**Worked comparison: training grants and the scrap rate**

Using Holzer et al. (1993) data on 54 firms reporting scrap rates in 1987, 1988 and 1989. The scrap rate is defective items per 100 produced, so a fall means higher productivity. GRANT indicates receipt of a 1988 job-training grant.

Both estimators say the same economic thing – a grant lowers the scrap rate, mostly with a one-year lag, on the order of 35–42%. FE is sharper because with T = 3 it keeps more information; FD agrees in sign and rough magnitude but pays for discarding 1987 with wider standard errors.

> **The intercepts mean different things**
>
> In the FE model, C = 0.49 is the normalised average firm intercept in levels.
>
> In the FD model, C = −0.14 is a constant in DIFFERENCES – a common downward trend in log scrap of about 14% per year across all firms. It is the differenced counterpart of a time effect, not a baseline scrap level.
>
> And the wildly different $R^2$ values are an artefact of what each specification explains, not evidence that one model is better. The near-zero FD $R^2$ says only that grant changes explain little of scrap changes.

# 5. FE-2SLS estimation

Fixed effects removes correlation with the time-invariant effect $a_i$. But a time-varying regressor can still be correlated with the idiosyncratic error, and FE does nothing about that.

$$log(scrap_{it}) = \beta_0 + \beta_1 hrsemp_{it} + a_i + \lambda_t + u_{it}$$

We might worry that hours of training per employee is correlated with $a_i$ – unmeasured fixed firm characteristics such as management quality or capital. Fixed effects handles that. But a firm facing a good productivity shock this year might simultaneously hire skilled workers and cut training, which makes hrsemp correlated with $u_{it}$ even after demeaning. Then we need an instrument for demeaned hrsemp.

## 5.1 First stage

$$hrsemp_{it} = \pi_0 + \pi_1 grant_{it} + \pi_2 grant_1_{it} + c_i + \tau_t$$

| Condition | Argument |
|---|---|
| Exogeneity | Michigan awarded grants roughly first-come, first-served, not on firm performance, so grant is plausibly uncorrelated with the firm's scrap shocks. Note this is an assumption about the assignment mechanism – and under FE it is the strict form, since demeaning pulls in $u_i$ from every period |
| Relevance | Test $H_0$: $\pi_1 = \pi_2 = 0$. We want an F-statistic above 10 to reject weak instruments |

The grant programme did not exist in 1987, so every firm has grant = 0 that year and then switches to 0/1 in 1988–89. That built-in movement is exactly the within-firm variation FE-2SLS needs.

Results: the grant effect of about +32.8 hours has the right sign and is economically large; the lagged effect is irrelevant; the instruments are not useless, since the joint null is rejected. But the F-statistic is a little below 10, so the instruments are borderline-adequate rather than comfortably strong.

### 5.2 Second stage and diagnostics

With 48 firms over three years, the design cannot detect an effect of training hours on the scrap rate. Two diagnostics are worth separating carefully:

| Diagnostic | What it tests | Result here |
|---|---|---|
| First-stage F | Instrument STRENGTH – are the instruments correlated enough with the endogenous regressor? | F just under 10 – borderline |
| Overidentification (J-statistic) | Instrument VALIDITY – are the instruments exogenous? With 2 instruments for 1 endogenous regressor there is q = 1 overidentifying restriction | Rejects exogeneity at 10% but not at 5% – validity is questionable |

*These answer different questions and should never be conflated. A strong instrument can be invalid, and a valid instrument can be weak.*

One technical caveat on this example: two-way FE plus clustering by 48 firms produces more effect parameters than the clustered covariance can support at full rank. It does not bias the coefficients, but it means the standard errors and the J-statistic should not be trusted to many decimal places. This is a mechanics demonstration rather than a precise estimation.

## 6. Random effects estimation

If we think $a_i$ is uncorrelated with each regressor in all periods, then transforming it away is wasteful – it discards information and yields inefficient estimators. The model becomes a random effects model under:

$$Cov(x_{itj}, a_i) = 0 \quad for\ all\ t\ and\ j$$

The difference from fixed effects is that the unit constants are treated as random rather than fixed parameters. $a_i$ is drawn once per unit from a distribution with variance $\sigma^2_a$ – you never learn its value, and across units the values scatter – but once drawn, that same number is carried into every period for that unit. It is not re-drawn each year. The random effects estimator is feasible GLS.

### 6.1 Comparing all three

| | Pooled OLS | Fixed effects | Random effects |
|---|---|---|---|
| Treats $a_i$ as… | part of the error | a parameter (swept out) | a random component |
| Key assumption | Cov(x, $a_i$) = 0 | none on $a_i$ | Cov(x, $a_i$) = 0 |
| If Cov(x, $a_i$) ≠ 0 | inconsistent | consistent | inconsistent |
| Variation used | within + between | within only | within + between (weighted) |
| Time-invariant regressors | estimated | dropped | estimated |
| Efficiency if RE holds | inefficient | consistent, not efficient | efficient (BLUE) |
| EViews setting | Cross-section: None | Cross-section: Fixed | Cross-section: Random |

*The third row is the one that matters. Only FE stays consistent when the unobserved effect is correlated with the regressors – which is the reason FE is the default when in doubt, and the reason we test.*

### 6.2 The wage equation again

Pooled OLS and RE both permit educ, black and hispan, and give similar coefficients on them; all three drop out of FE. Both the marriage and union premiums fall notably moving from pooled OLS to RE, and fall again moving to FE.

> **Why RE gives bigger premiums than FE**
>
> RE assumes $a_i$ is uncorrelated with the regressors. If that is false – if union men have stable unobserved characteristics such as steady productivity or motivation – then the estimated premium absorbs part of those characteristics.
>
> The FE estimate is cleaner: it identifies the premium purely from men who change status.
>
> Note also that EViews' "Unweighted Statistics" for an RE model are computed from the original untransformed data, not from the quasi-demeaned data GLS actually estimates on.

### 6.3 The Hausman test

$$H_0: Cov(x, a_i) = 0 \quad versus \quad H_1: Cov(x, a_i) \neq 0$$

Under the null both FE and RE are consistent but FE is inefficient. Under the alternative FE remains consistent while RE does not. The statistic may be viewed as a distance measure between the two estimators. Reject $H_0$ and RE is inconsistent, so use fixed effects.

In the wage example we reject. The comparison table shows why: MARRIED falls from 0.064 to 0.047 and UNION from 0.106 to 0.080 moving from RE to FE. That gap between the columns IS the correlation between $a_i$ and the regressors – the endogeneity RE assumes away but the data reject.

Note that only three variables enter the test (df = 3), because Hausman can only compare coefficients that FE actually estimates. EDUC, BLACK and HISP are swept out, and EXPER is collinear with the period effects.

RE does have advantages: fewer parameters to estimate, and it recovers coefficients on time-invariant regressors that FE cannot. But if the unobserved effects are correlated with the regressors, RE is biased and inconsistent – and in many applications the whole reason for using panel data is precisely to allow that correlation.

## 7. Diagnostics and choosing an estimator

### 7.1 Four things to check

| Check | What it tells you |
|---|---|
| Is there an unobserved effect at all? | Breusch-Pagan LM test, $H_0$: $\sigma^2_a = 0$. Reject → heterogeneity matters, do not stop at pooling |

| Check | What it tells you |
|---|---|
| Serial correlation | $a_i$ induces within-unit correlation by construction even under the RE assumption, and $u_{it}$ may itself be autocorrelated. Inspect residuals or use a Wooldridge-type test |
| Heteroskedasticity | Error variance differs across units – large versus small firms, rich versus poor countries. Estimators stay consistent but become inefficient, and the standard errors are invalid, so conclusions from t-statistics may be wrong |
| Cross-sectional dependence | Units correlated at a point in time through common shocks. Mostly a concern in macro panels with sizeable T |

## 7.2 The workhorse fix: cluster-robust standard errors

One correction handles both heteroskedasticity across units and arbitrary serial correlation within a unit. Point estimates are unchanged; only inference is repaired. Use it by default with panel data.

> **The EViews naming trap**
>
> Coefficient covariance → White PERIOD ✓ clusters within a cross-section over time, i.e. entity clustering. This is what you want.
>
> White CROSS-SECTION ✗ clusters contemporaneously across units, by period – the opposite.
>
> The names are the reverse of what intuition suggests, and choosing the wrong one silently gives you the wrong standard errors.

Two habits worth keeping: include time dummies by default, setting period effects to Fixed to absorb aggregate shocks and trends common to all units – cheap insurance Wooldridge recommends routinely. And match the transformation to the error: if the idiosyncratic error looks like a random walk, prefer first differencing to FE.

## 7.3 A decision workflow

1. Add time dummies to absorb common shocks from the start.
2. Test for $a_i$ with the LM test after fixed effects estimation – is pooling adequate, or does heterogeneity matter?
3. Choose between FE and RE with the Hausman test, plus economic intuition. Reject → FE, since only FE is consistent when correlation is present. Fail to reject → RE, which is consistent and efficient under that null and recovers the coefficients on time-invariant regressors that FE cannot.
4. Cluster by entity. Always.

## 7.4 What lies beyond this course

| Topic | Issue |
|---|---|
| Dynamic panels | A lagged dependent variable makes FE biased – Nickell bias – handled by GMM estimators such as Arellano-Bond. The bias shrinks as T grows: negligible once $T \geq 20$–30, non-trivial when $T \leq 10$, and worst at T = 2–3 |

| Topic | Issue |
|---|---|
| Large-T heterogeneous panels | Mean Group and Pooled Mean Group estimators, built on ARDL and cointegration – a macro-panel topic |
| Cross-sectional dependence | Common factors and spatial linkages; Pesaran CD test, spatial panels |
| Panel VARs and ECMs | VARs and SVARs can be estimated on panel data, with impulse response analysis |

# 8. Difference-in-differences

DiD estimates the impact of a treatment by comparing the before-and-after change in the outcome for those treated with the change for those not treated. It can be estimated on repeated cross sections, but most econometric work uses panel data.

With a single treatment date D*, treated group T and never-treated group NT:

$$\delta = (Y_{T,\, t \geq D^*} - Y_{NT,\, t \geq D^*}) - (Y_{T,\, t < D^*} - Y_{NT,\, t < D^*})$$

δ is the Average Treatment Effect for the Treated (ATET).

## 8.1 DiD as a two-way fixed effects regression

$$Y_{i,g,t} = \delta\, D_{g,t} + a_i + \lambda_t + u_{i,g,t}$$

where g identifies the treatment (g = 1) and control (g = 0) groups, and D takes the value 1 whenever a treatment-group observation is actually treated. The ATET equals δ.

Since this is estimated by OLS, every OLS assumption applies. But DiD needs one more:

> **The parallel trends assumption**
>
> In the counterfactual where the treated group is not treated, the difference between treatment and control groups is constant through time.
>
> Without it, $\hat{\delta}$ is not unbiased for the ATET – it picks up pre-existing divergence in trends as if it were a treatment effect.
>
> Multiple treatment dates do not change the form of the estimator: simply extend the two-way FE model to more than two groups.

## 8.2 Minimum wages and teen employment

Callaway and Sant'Anna (2021) study county-level teen employment from 2003 to 2007. The federal minimum wage stayed at $5.15 throughout, but some states raised theirs above the federal floor. Counties in those states are treated, with different treatment times; counties in states that did not change are controls.

| Result | Value |
|---|---|
| Effect on county teen employment | TREATED = −0.0365, so about −3.7%, statistically significant (p ≈ 0.028) |
| Parallel trends | Not rejected (p = 0.80) – treated and control counties were not on significantly different paths before the increases |

In EViews the DiD method is exactly panel least squares with cross-section and period fixed effects and White-period standard errors – the built-in DiD object and a manual two-way FE estimation reproduce each other identically, down to the coefficient, standard error, SSR and $R^2$. The DiD interface simply adds post-estimation diagnostics.

### 8.3 Other considerations

Additional exogenous regressors can be included without changing the behaviour of the DiD estimator. But recent research notes that two-way FE is unsuitable for multiple-timing DiD when the treatment effect changes as time from treatment increases. Three responses, all available in EViews under View → Difference-in-Difference Diagnostics:

- Group-Time Average Treatment Effects (Callaway-Sant'Anna) – robust to the effect changing with time since treatment.
- Goodman-Bacon decomposition – calculates treatment effects for individual pairs of treatment groups.
- Borusyak, Jaravel and Spiess imputed estimator.

## Appendix. The correlated random effects approach

Where it makes sense to view $a_i$ as random, there is an alternative to fixed effects that still allows $a_i$ to be correlated with the regressors. Run RE, but add the time-averages of the time-varying regressors as extra controls:

$$y_{it} = \alpha + \beta\, x_{it} + \gamma\, \bar{x}_i + r_i + u_{it}, \qquad \text{where } a_i = \alpha + \gamma\, \bar{x}_i + r_i$$

Including the within-unit mean $\bar{x}_i$ controls for the correlation between $a_i$ and $x_{it}$; what is left over, $r_i$, is uncorrelated with the regressors. The specification implies $a_i$ and $x_{it}$ are correlated whenever $\gamma \neq 0$.

| Property | Why it is useful |
|---|---|
| β equals the FE estimate | The slope on $x_{it}$ is numerically identical to fixed effects. One equation displays the FE-RE link |
| The Hausman test is a Wald test | A joint test of $H_0$: γ = 0 on the time-averages is the Hausman test – same null, but an ordinary cluster-robust Wald test |
| γ names the culprits | A significant γ on a particular $\bar{x}_i$ tells you WHICH regressor is correlated with $a_i$, which the Hausman statistic alone cannot |
| Time-constant regressors allowed | CRE provides a way to include them in what is effectively a fixed effects analysis |

An example: if $x_{it}$ is the tax rate on firm profits in county i in year t and $y_{it}$ is county output, including the average tax rate over the T years allows for systematic differences between historically high-tax and low-tax counties – differences that may themselves affect output.

#### Doing it in EViews

```
' time-average of concen, by route
series concenbar = @meansby(concen, @crossid)
' random-effects equation with the mean added
```

```
equation eq_cre.ls(cx=r, cov=perwhite) log(fare) c concen concenbar
' robust Hausman: joint Wald test on the mean
eq_cre.wald c(3)=0
```

Set the coefficient covariance to White period (entity-clustered) so the Wald test is robust. Check that the coefficient on concen equals the fixed-effects estimate exactly. The data are Wooldridge's AIRFARE panel: 1,149 US air routes over 1997–2000, with average fare, market concentration, distance and endpoint populations.

## Key terms

| Term | Meaning |
|---|---|
| Panel (longitudinal) data | The same N units observed over T periods, giving N × T observations. |
| Balanced / unbalanced panel | Every unit observed in every period / or not. |
| Unobserved effect $a_i$ | Time-constant unobserved factors specific to a unit. Also called individual heterogeneity or the fixed effect. |
| Idiosyncratic error $u_{it}$ | Unobserved factors varying over both i and t. |
| Composite error $v_{it}$ | $a_i + u_{it}$ – what pooled OLS leaves in the error. |
| Two-way error components | $v_{it} = a_i + \lambda_t + u_{it}$, adding a time effect common to all units in a period. |
| Pooled OLS | Ignores $a_i$. Consistent only if $Cov(x, a_i) = 0$; errors serially correlated by construction. |
| Within (fixed effects) estimator | OLS on time-demeaned data. Allows arbitrary $Cov(x, a_i)$; sweeps out time-constant regressors. |
| LSDV | Least squares with a dummy per unit. Identical results to demeaning; infeasible for large N. |
| First differencing | Removes $a_i$ by subtracting the previous period. Loses one period; preferable when $u_{it}$ is a random walk. |
| Random effects | Feasible GLS treating $a_i$ as a random draw. Efficient if $Cov(x, a_i) = 0$, inconsistent otherwise. |
| Hausman test | $H_0$: $Cov(x, a_i) = 0$. Reject → use fixed effects. |
| FE-2SLS | Fixed effects combined with instrumental variables, for a time-varying regressor correlated with $u_{it}$. |
| Cluster-robust standard errors | Clustered by entity; repairs heteroskedasticity and within-unit serial correlation at once. "White period" in EViews. |
| Nickell bias | The bias in FE when a lagged dependent variable is included; shrinks as T grows. |
| ATET | Average Treatment Effect for the Treated – δ in the DiD regression. |
| Parallel trends | Absent treatment, the gap between treated and control groups would be constant over time. |
| Correlated random effects (CRE) | RE plus the within-unit means of the regressors; reproduces the FE slope and makes Hausman a Wald test. |

## Review questions

Use these to check your understanding before the seminar.

1. What distinguishes a panel from a pooled cross section? Give an example of each.
2. Name the single biggest advantage of panel data and explain the mechanism by which it works.
3. Write the unobserved effects model and identify $a_i$, $u_{it}$ and $v_{it}$. Which one has no t subscript, and why does that matter?
4. Distinguish one-way from two-way error components. Give an economic example of $\lambda_t$.
5. Pooled OLS is sometimes consistent. Under what condition, and why are its standard errors still wrong?
6. Derive the within transformation and show that $a_i$ cancels.
7. Why can gender not be included in a fixed effects wage regression?
8. Why does EXPER drop out of the FE model while EXPER² survives?
9. MARRIED falls from 0.108 to 0.047 moving from pooled OLS to FE. Explain economically what the FE estimate is using that pooled OLS was not.
10. The FE model reports $R^2 = 0.62$ against 0.19 for pooled OLS. Why is "three times the fit" the wrong reading?
11. EViews reports a single intercept for an FE model. What is it, and how would you recover unit i's own intercept?
12. Cross-section F = 9.16 and period F = 29.24, both rejecting. What do you conclude about the specification?
13. Fixed effects removes $a_i$. Give a concrete example where a time-varying regressor is still endogenous, and say what is needed.
14. Distinguish the first-stage F-test from the overidentification test. What does each test, and what did each conclude in the training example?
15. Why does the grant programme not existing in 1987 help identification?
16. Explain what "$a_i$ is drawn once per unit" means in the RE model. Is it re-drawn each period?
17. Using the three-estimator table, explain in one sentence why FE is the default when in doubt.
18. Why are the RE union and marriage premiums larger than the FE ones? What does the gap between them measure?
19. State the Hausman null and both hypotheses. Why does the test have only 3 degrees of freedom in the wage example?
20. In EViews, which covariance option clusters by entity? What does the other one do, and why is the naming a trap?
21. Set out the four-step decision workflow. What would make you stop at random effects?
22. What is Nickell bias, and at which panel dimensions is it worst?
23. Write the DiD estimator in terms of four group means. What does each of the two differences remove?

24. State the parallel trends assumption. What goes wrong if it fails, and how was it checked in the minimum wage example?
25. The built-in DiD object and a manual two-way FE regression give identical output. What does this tell you about what DiD actually is?
26. In the CRE regression, what does a significant $\gamma$ on $\bar{x}_i$ tell you that the Hausman statistic alone does not?
27. Why does the coefficient on $x_{it}$ in a CRE regression equal the FE estimate exactly?

## References and further reading

Wooldridge, J. M. Introductory Econometrics: A Modern Approach – Chapters 13 and 14 cover panel data methods.

Asteriou, D., Hall, S.G. Applied Econometrics – Chapter 21.